

AI Engineer Intern - Technical Assessment

You will be building a weather app that will take in user-input and then gather data from outside sources (API retrieval) to provide the user with relevant real-time information based on their requests.

Each assessment will be looking not-only for your technical aptitude but also your ability to think creatively and design innovative solutions. We want you to think like a user who needs to know the weather in their current location or another location of interest they hope to be traveling to. What are some things they should consider that might not be obvious?

This technical assessment is split into two sections:

- **Frontend Engineers** should complete Tech Assessment **#1**. You will be responsible for what users see and how they interact on a website or application. *(Do not use Python or Java frameworks for frontend assignment)*
 - ▶ Does your final submission adapt seamlessly to various screen sizes and devices (desktops, tablets, smartphones)?
 - ▶ What responsive design techniques were used?
 - ▶ Are you able to manage a few APIs? *(most API management will be handled by backend during the internship)*
- **Backend Engineers** should complete Tech Assessment **#2**. You will be responsible for the server-side applications as well as API and database management
 - ▶ Are you able to perform CRUD (Create, Read, Update, Delete) operations based on weather and location information. *Feel free to use either SQL or NoSQL database.*
 - ▶ Do you have the ability to design and build RESTful APIs *(or other API architectures)* for communication between the front-end and back-end, or between different back-end services?
- **Full Stack Engineers** = If you have the ability, we highly encourage completing both Tech Assessment **#1 and #2**. *(This candidate will have the highest priority) (Do not use Python or Java frameworks for frontend assignment)*

The ideal candidate will combine their technical skills to guide the user through their input, validate their input and then store their input and retrieved API data into a database.

Please let us know which assessment has been completed.

Tech assessment 1: Weather App *(frontend engineers)*

What We're Looking For:

Make a weather app that:

- Let users enter a location (*any of the following: Zip Code/Postal Code, GPS Coordinates, Landmarks, Town, City, etc..*) and get the current weather. You have autonomy to determine how to handle how the user should enter their location
- Shows the weather clearly, with any details you think are useful.
- Let users see the weather based on their current location.
- Use icons or images to make the weather info look cool using any design standards.

To Stand Apart:

- Add a 5-day forecast as stated in **1.1 Five-Day Forecast** section (*below*)
- Add error-handling functionality as stated in **1.2 Error Handling** section (*below*)

Technology Stack = Any **Javascript** (*ReactiveJS, NextJS, Angular, Vue.js, SolidJS*) frameworks for this front-end portion. (*Do not use Python or Java frameworks for frontend assignment*). Also focus on web first approach (not mobile)

1.1 Five-Day Forecast - Display the 5-day forecast in a clear and organized manner. This could be a horizontal or vertical list, a grid, or any other user-friendly layout.

1.2 Error Handling - Show an example or two of handling errors gracefully (e.g., display a message if the city is not found or if the API request fails).

Note: Please don't give static information. Use any **APIs** to pull real weather information.

NO NEED to make the UI beautiful. Our designer will make your AI product pretty. We just want to see your technical skills. However, if you want to make it nice, feel free to do it. It's nice to have.

What We Need From You: See [this section](#) below which is the same for Tech Assessment 1 & 2.

Tech assessment 2: Weather App (backend engineers)

What We're Looking For:

This section will focus on persistence with CRUD functionality as well as API calls and error handling. The more functionalities are in your code the greater the credibility of your submission.

Make a weather app that:

- Let users enter a location (*any of the following: Zip Code/Postal Code, GPS Coordinates, Landmarks, Town, City, etc..*) and get the current weather. You have autonomy to determine how to handle how the user should enter their location
- Shows the weather clearly, with any details you think are useful.
- Show database skills based on the **2.1 CRUD** requirements (*below*)
- Show an example of exporting data to any format as described in **2.3 Export Data** (*below*)

To Stand Apart:

- Perform additional API integration as described in section **2.2 API Integration** (*below*).

2.1 CRUD - Please choose a database (SQL or noSQL) to have data persistence and incorporate the following functionality.

- **CREATE** - Allow user(s) to enter both a location and a date range and output temperature for that location within that date ranges specified. Store all this information into a database.
 - Validate the date ranges
 - Validate that the location really exists (or allow fuzzy match for system to determine location)
- **READ** - Allow users to also be able to read any of the weather information (and anything else stored in the database) they have requested previously (or even what others have entered). - Row level security is not necessary to segment the data by users.
- **UPDATE** - Allow users to be able to update any of the weather information stored in the database. (You can pick which data should and shouldn't be updated).
 - As above have validations so user input is not incoherent or a location is incorrect.
- **DELETE** - Allow users to be able to delete any of the records in the database.

2.2 API Integration - Based on the information users provided (above) perform any of the following:

- Provide Youtube videos of the location user chose
- Provide Google Map data of the location. *(If location is not specific it is ok to show an approximate point of interest)*
- Be creative by using any additional API's.
 - Here are some free API's for reference = <https://www.freecodecamp.org/news/public-apis-for-developers/>

2.3 Data Export - Allow users to export data from the database into JSON, XML, CSV (delimited), PDF, Markdown output format(s). - Feel free to pick whichever (or all) format(s) you prefer for the export.

What We Need From You *(for both Tech Assessment #1 & #2)*

- **When:** Please send us your project AS SOON AS POSSIBLE by submitting this [Google form](#) no later than 7 days after receiving the assessment. If for special scenarios, you need more time to complete it, let us know. (If Dual Role applicant, then have 10 days to complete)
- **How:** Share a link to your **GitHub repo** with the code and a brief README on how to run it and what you did.
 - IMPORTANT: Public vs. Private repository options
 - **Set your Github repository “public” and “open-source”.** Or our tech team can't access and evaluate your code. You can make the repository “private” after we finish code evaluation.
 - Alternatively, you can set your repository to **private** and then must allow collaboration from Github account = community@pmaccelerator.io & hr@pmaccelerator.io
 - Allow the ability to **Clone or Download**
 - Have a **requirements file** show which libraries/packages need to be installed
- Share a short **demo video** explanation of your project. Provide a recorded screen sharing and also 1-2min demo showing & explaining both the code and output results based on above requirements. Share a viewable URL of the demo video in your submission. *(can be hosted on Google drive or YT, or Vimeo, etc..)*

- The weather app should include your name and an informational details which provides a description of the PM Accelerator (*our LinkedIn page: Product Manager Accelerator has our company's description*).